

Introduction to Topology

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Abstract

This course was taught by Dr. Yu-Shen Lin at Boston University during the Spring 2025 semester. We used the text Introduction to Topology, Pure and Applied by Colin Adams and Robert Franzosa.

Contents

1	Topological Spaces
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2

Chapter 1

Topological Spaces

Lecture 1: Syllabus Day

Tue 21 Jan 2025 11:01

Textbook not required but I already bought it wooh also biweekly homework on blackboard always due at 11:59pm. We are encouraged to type homework (awesomewwww) also why does this one guy in the class look like a literal reddit mod LOL also karim showed up 7m late. “I cannot give everyone an A” great its competitive now

“You’ve probably heard that a mug and a donut are the same thing” I love this class. Making this notion precise and rigorous is difficult, and by the end of the semester we will be able to prove this notion.

Definition 1.1: Topological Space

Let $X \neq \emptyset$ be a set. A topology on X is a collection of subsets $\mathcal{T} \subseteq \mathcal{P}(X)$ such that

1. $\emptyset, X \in \mathcal{T}$;
2. $\mathcal{T}$ is closed under *finite* intersection;
3. $\mathcal{T}$ is closed under union.

We call the pair $(X, \mathcal{T})$ a *topological space*.

Definition 1.2: Open Set

Let $(X, \mathcal{T})$ be a topological space. Any subset $A \subseteq X$ such that $A \in \mathcal{T}$ is called an *open set*.

Note that there is a finite restriction on (2), but not on (3). We will discuss this later. Definition 1.1 is extremely abstract and seemingly arbitrary, so consider the following examples to build some intuition.

The most basic example is that of the *standard topology on $\mathbb{R}$* . Define $\mathcal{T}$ as the collection of all unions of open intervals of the form $(a, b) \subseteq \mathbb{R}$, together with the empty set. Trivially this definition satisfies property (1), since $(-\infty, \infty) = \mathbb{R}$ is open, and we define $\emptyset \in \mathcal{T}$. It also satisfies (3) by our definition. Showing (2) is slightly more difficult and requires ε - δ to make precise, and we will cover it later when we have this machinery.

Consider the open sets of the form

$$\left(-\frac{1}{2^n}, \frac{1}{2^n}\right), \quad n \in \mathbb{N}$$

in the standard topology on $\mathbb{R}$. However, if we take the intersection

$$\bigcap_{n \in \mathbb{N}} \left(-\frac{1}{2^n}, \frac{1}{2^n}\right) = \{0\}.$$

This is *not an open set* in the standard topology on $\mathbb{R}$, as it is not an open interval and not empty. This is why the finite criterion on (2) is important.

Consider a more simple example. On any set X , the trivial topology $\mathcal{T} = \{\emptyset, X\}$ is a topology. We trivially have criteria (1). Consider a finite collection $\{A_n\}_{n=0}^k$. There are two cases. If $A_n = \emptyset$ for some n , then

$$\bigcap_{n=0}^k A_n = \emptyset.$$

Otherwise, we have

$$\bigcap_{n=0}^k A_n = X,$$

since we must then have $A_n = X \forall n$. Since $\emptyset, X \in \mathcal{T}$, we have (2). The same argument applies for (3); if $X \in \{A_n\}_{n=0}^k$, we have the union is X ; otherwise we have the union is empty. The trivial topology is also called the minimal topology, as it is the smallest possible topology since it must contain these subsets.

The next example is the discrete topology. Let $\mathcal{T} = \mathcal{P}(X)$ for some set X . This very trivially satisfies definition 1.1, and we will explain later why we call it the discrete topology. This is also called the maximal topology, since it is the largest possible topology.

Let $X = \mathbb{R}$. We will now define what is called the Zariski topology, or sometimes the finite compliment topology. Let $\mathcal{T}$ be defined as all sets with either finite compliment in $\mathbb{R}$, or the empty set:

$$\mathcal{T} := \{Y \subseteq \mathbb{R} \mid Y = \emptyset \vee |\mathbb{R} \setminus Y| < \infty\}.$$

We will show this is a topology. By definition, $\emptyset \in \mathcal{T}$, and we have $\mathbb{R} \setminus \mathbb{R} = \emptyset$ is finite, so $\mathbb{R} \in \mathcal{T}$. The second example is more interesting. Consider a finite collection $\{A_n\}_{n=1}^k \subseteq \mathcal{T}$. We want to show

$$\bigcap_{i=1}^k A_i$$

is open. By our definition, a set is open if and only if it's compliment in $\mathbb{R}$ is finite. DeMorgan's Law states that

$$\left(\bigcap_{i=1}^k A_i\right)^c = \bigcup_{i=1}^k A_i^c.$$

"I assume you learned this in high school". Note that the maximum cardinality of this union occurs when $A_i \cap A_j = \emptyset$ for all $i \neq j$, so we have

$$\left|\bigcup_{i=1}^k A_i^c\right| = \sum_{i=1}^k |A_i^c|.$$

A finite sum of finite numbers is finite. More generally, a finite union of finite sets is finite. Now we show the third criterion holds. Suppose $\{A_i\}_{i \in I} \subseteq \mathcal{T}$. By definition, $|A_i^c| < \infty$ for all $i \in I$. We wish to show

$$\left(\bigcup_{i \in I} A_i\right)^c < \infty.$$

DeMorgan's Law also states that

$$\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c.$$

Since the intersection of finite sets is finite, we have that $\bigcup \{A_i\}_{i \in I}$ is open. The Zariski topology is the topology used in algebraic geometry.

Definition 1.3: Neighborhood

Let X be a topological space and let $x \in X$. An open set U such that $x \in U$ is called a *neighborhood* of x .

The intuition behind a neighborhood is the collection of points near x . This may not be true in a general topology, but it is true in many topologies.

Theorem 1.1

Let X be a topological space. Then $A \subseteq X$ is open if and only if for all $x \in A$, there exists a neighborhood U such that $x \in U \subseteq A$.

Proof. Suppose A is open and let $x \in A$. Then $U = A$ is a neighborhood of x . Now suppose for all $x \in A$, there exists an open neighborhood $U_x \subseteq A$ of x . Note that

$$A \subseteq \bigcup_{x \in A} U_x$$

because

$$y \in A \implies \exists U_y \text{ s.t. } y \in U_y \implies y \in \bigcup_{x \in A} U_x.$$

However, since every $U_x \subseteq A$, we have also that

$$\bigcup_{x \in A} U_x \subseteq A$$

since

$$y \in \bigcup_{x \in A} U_x \implies \exists U_x \text{ s.t. } y \in U_x \subseteq A.$$

Therefore, A is open by criteria (3) in definition 1.1. ■

Lecture 2: hi

Thu 23 Jan 2025 10:58

I walked in and we're talking about some stuff with boats and physics? I don't really know what's happening, but we did some vector stuff to conclude the best way to sail against the wind is at a 30 degree angle.

It is usually too difficult to list all open sets in a given topology, so we use a basis instead.

Definition 1.4: Basis

Let X be a nonempty set and let $\mathcal{B}$ be a collection of subsets of X . Then $\mathcal{B}$ is a *basis* of some topology $\mathcal{T}_{\mathcal{B}}$ on X iff

1. For all $x \in X$, there exists $V \in \mathcal{B}$ such that $x \in V$;
2. For all $V_1, V_2 \in \mathcal{B}$ with $x \in V_1 \cap V_2$, there exists $V_3 \in \mathcal{B}$ such that $x \in V_3 \subseteq V_1 \cap V_2$.

Proposition 1.1

Let $\mathcal{B}$ be a basis for some set X . Then the set $\mathcal{T}_{\mathcal{B}}$ of the empty set, together with all unions of elements of $\mathcal{B}$, forms a topology on X .

Proof. Recall the three axioms from definition 1.1. By definition, $\emptyset \in \mathcal{T}_{\mathcal{B}}$. By property (1) of definition 1.4, for all $x \in X$, there exists $V_x \in \mathcal{B}$ such that $x \in V_x$. By definition of $\mathcal{T}_{\mathcal{B}}$,

$$X = \bigcup_{x \in X} V_x \in \mathcal{T}_{\mathcal{B}}.$$

Therefore, property (1) is satisfied.

Now, we show property (2); closure under finite intersection. Let $\{A_i\}_{i=1}^k \subseteq \mathcal{T}_{\mathcal{B}}$. By definition, we have that A_i is either empty, or can be represented as a union

$$A_i = \bigcup_{j \in J_i} V_{ij}$$

for $\{V_{ij}\}_{j \in J_i} \subseteq \mathcal{B}$. If any $A_i = \emptyset$, we have that the intersection

$$\bigcap_{i=1}^k A_i = \emptyset \in \mathcal{T}_{\mathcal{B}}.$$

Thus, we can assume WLOG that $A_i \neq \emptyset$ for all i . We have

$$\bigcap_{i=1}^k A_i = \bigcap_{i=1}^k \bigcup_{j \in J_i} V_{ij}.$$

Recall that the distributive property says $a \cup (b \cap c) = (a \cup b) \cap (a \cup c)$. We can thus rewrite something along the lines of

$$\bigcap_{i=1}^k \bigcup_{j \in J_i} V_{ij} = \bigcup_{j_i \in J_i \text{ } i \in I} \bigcap_{i \in I} V_{ij_i}.$$

We thus need only prove this intersection is an element of $\mathcal{T}_{\mathcal{B}}$. We prove this by strong induction. For $k = 1$, we have $V_1 = \bigcap_{i=1}^1 B_i \in \mathcal{B} \subseteq \mathcal{T}_{\mathcal{B}}$. Now assuming the inductive hypothesis for $n \leq k$, we have for any $\{V_i\}_{i=1}^{k+1} \subseteq \mathcal{B}$, the intersection

$$\bigcap_{i=1}^k V_i \in \mathcal{B}.$$

We can thus represent it as a union of elements $\{V'_i\}_{i \in I} \in \mathcal{T}_{\mathcal{B}}$. By distributivity, it follows that

$$\bigcap_{i=1}^{k+1} V_i = \left(\bigcup_{i \in I} V'_i \right) \cap V_{k+1} = \bigcup_{i \in I} (V'_i \cap V_{k+1}).$$

By definition 1.4, for all i and for all $x \in V'_i \cap V_{k+1}$, there exists $V_x \subseteq V'_i \cap V_{k+1}$ such that $x \in V_x$. We then have

$$\bigcup_{i \in I} (V'_i \cap V_{k+1}) = \bigcup_{i \in I} \bigcup_{x \in V'_i \cap V_{k+1}} V_x.$$

Since $V_x \in \mathcal{B}$ for all x , by definition of $\mathcal{T}_{\mathcal{B}}$, we have

$$\bigcup_{i \in I} \bigcup_{x \in V'_i \cap V_{k+1}} V_x \in \mathcal{T}_{\mathcal{B}}.$$

By induction, we are done.

Finally, we show property (3). Let $\{A_i\}_{i \in I} \subseteq \mathcal{T}_{\mathcal{B}}$. By definition of $\mathcal{T}_{\mathcal{B}}$, for all i there exists $\{B_{ij}\}_{j \in J_i}$ such that

$$A_i = \bigcup_{j \in J_i} B_{ij}.$$

We then have

$$\bigcup_{i \in I} A_i = \bigcup_{i \in I} \bigcup_{j \in J_i} B_{ij} = \bigcup_{i \in I, j \in J_i} B_{ij}.$$

Since this is a union of elements in $\mathcal{B}$, by definition of $\mathcal{T}_{\mathcal{B}}$, we have that

$$\bigcup_{i \in I, j \in J_i} B_{ij} \in \mathcal{T}_{\mathcal{B}}.$$

■

Definition 1.5: Topology Generated by a Basis

Let $\mathcal{B}$ be a basis for a set X . Then the set $\mathcal{T}_{\mathcal{B}}$ consisting of the empty set together with all unions of elements in $\mathcal{B}$ is a topology on X , called the *topology generated by $\mathcal{B}$* .

For example, define

$$B_r(x) := \{a \in \mathbb{R} \mid |x - a| < r\}.$$

Then the collection

$$\mathcal{B} = \{B_r(x) \mid x \in \mathbb{R}, r \in \mathbb{R}^+\}$$

is a basis for the standard topology on $\mathbb{R}$. This generalizes; let

$$B_r(x) := \{a \in \mathbb{R}^n \mid \|x - a\| < r\}.$$

Then

$$\mathcal{B} = \{B_r(x) \mid r \in \mathbb{R}^+, x \in \mathbb{R}^n\}$$

is a basis for $\mathbb{R}^n$. We will prove this. Let $x \in \mathbb{R}^n$. Note that for any choice $r \in \mathbb{R}^+$, we have $x \in B_r(x)$. This shows property (1) of definition 1.4. As for property (2), let $x \in B_{r_2}(x_1) \cap B_{r_2}(x_2)$. It follows that this intersection is nonempty, which implies that $\|x_1 - x\|, \|x_2 - x\| < r_1, r_2$. We wish to find some r such that $B_r(x) \subseteq B_{r_1}(x_1) \cap B_{r_2}(x_2)$. I propose we choose $|r_1 - r_2|$. Let $y \in B_{|r_1 - r_2|}(x)$. We wish to show $\|x_1 - y\| < r_1$ and $\|x_2 - y\| < r_2$. We have by the triangle inequality that

$$\|y - x_1\| \leq \|y - x\| + \|x - x_1\|.$$

We thus need only show this for $\|y - x_1\| = \|y - x\| + \|x - x_1\|$. Note that

$$\|y - x\| < |r_1 - r_2|, \quad \|x - x_1\| < r_1.$$

We thus have

$$\|y - x_1\| < |r_1 - r_2| + r_1 = |r_2 - r_1 + r_1| = r_1,$$

because $r_1 > 0$. Therefore, $\|y - x_1\| < r_1$. The proof is equivalent for r_2 . Therefore, $B_{|r_1 - r_2|}(x) \subseteq B_{r_1}(x_1) \cap B_{r_2}(x_2)$, and thus by definition 1.4, $\mathcal{B}$ is a basis for $\mathbb{R}^n$.

Note that the only nontrivial thing used here is the triangle inequality. In the future, we will define a *metric*, which is an abstraction of distance which maintains this, and a few other important properties of distance. We will then find that the collection of all open balls where the norm is replaced by the metric forms a basis for the metric space. This will be called the metric topology.

Note that the basis of a topology is not unique.

Lecture 3: Bases, Closed sets, Hausdorff

Tue 28 Jan 2025 11:05

Proposition 1.2

Let $(X, \mathcal{T})$ be a topological space and let $\mathcal{C}$ be a collection of open subsets of X . Then $\mathcal{C}$ is a basis for the topology $\mathcal{T}$ on X if and only if for all open sets $U \in \mathcal{T}$, for all $x \in U$, there exists $V \in \mathcal{C}$ such that $x \in V \subseteq U$.

Proof. For all $U \in \mathcal{T}$ and for all $x \in U$, let there exist $V \subseteq U$ in $\mathcal{C}$ such that $x \in V$. Since X is open, we have that for all $x \in X$, there exists $V \in \mathcal{C}$ such that $x \in V$. Now let $V_1, V_2 \in \mathcal{C}$. Since $\mathcal{C}$ is a collection of open sets, $V_1 \cap V_2$ is open. We thus have that for all $x \in V_1 \cap V_2$, there exists $V_3 \subseteq V_1 \cap V_2$ in $\mathcal{C}$ such that $x \in V_3$. By definition 1.4, $\mathcal{C}$ is a basis. We now need to show $\mathcal{T}_{\mathcal{C}} = \mathcal{T}$. Let $U \in \mathcal{T}$. For all $x \in U$, there exists $V_x \in \mathcal{C}$ such that $x \in V_x \subseteq U$. We then have

$$U = \bigcup_{x \in U} V_x.$$

$V_x \in \mathcal{C}$, $U \in \mathcal{T}_{\mathcal{C}}$. Additionally, since we defined $\mathcal{C}$ as a collection of *open* sets, and all $V \in \mathcal{T}_{\mathcal{C}}$ are unions of elements in $\mathcal{C}$, by definition 1.1 we have immediately that $\mathcal{T}_{\mathcal{C}} \subseteq \mathcal{T}$.

Let $\mathcal{C}$ be a basis for $\mathcal{T}$. Let $U \in \mathcal{T} = \mathcal{T}_{\mathcal{C}}$, and let $x \in U$. Since $\mathcal{C}$ is a basis, we can thus write

$$x \in U = \bigcup_{i \in I} V_i = U, \quad V_i \in \mathcal{C}.$$

Note that x is in this union. Therefore, there exists some $V_i \in \{V_i\}_{i \in I}$ such that $x \in V_i$, and we get for free that

$$V_i \subseteq \bigcup_{i \in I} V_i.$$

Therefore, for all open sets U , for all $x \in U$, there exists $V_i \in \mathcal{C}$ such that $x \in V_i \subseteq U$. ■

Definition 1.6: Closed

A subset $Y \subseteq X$ is *closed* in X if its complement $Y^c = X \setminus Y$ is open.

For example, in the standard topology on $\mathbb{R}$, a closed interval $[a, b]$ with $a < b$ is closed, because $[a, b]^c = (-\infty, a) \cup (b, \infty)$, and since both of these are open intervals, this set is open.

Lemma 1.1

$$U \in \mathcal{T} \iff \forall x \in U, \exists V_x \in \mathcal{B} \text{ s.t. } x \in V_x \subseteq U.$$

Proof. Follows from previous prp. ■

Using lemma 1.1, we can show the same for $\mathbb{R}^2$. Let $\overline{B}_\varepsilon(x) = \{y \in \mathbb{R}^2 \mid d(y, x) \leq \varepsilon\}$. Let $x' \in \overline{B}(x, \varepsilon)^c$,

and note that $d(x, x') > \varepsilon$, so $d(x', x) - \varepsilon > 0$. Take

$$= 1/2\gamma = \frac{1}{2}(d(x', x) - \varepsilon) > 0.$$

By definition, $B(x', \gamma)$ is open, and contains x' . It thus suffices to show by lemma 1.1 that $B(x', \gamma) \subseteq X \setminus \overline{B}(x, \varepsilon)$. That is, for all $y \in B(x', \gamma)$, $d(x, y) > \varepsilon$. Note that $y \in B(x', \gamma)$ implies

$$d(x', y) < \gamma = \frac{1}{2}(d(x', x) - \varepsilon) < d(x, x') - \varepsilon.$$

We thus have by the triangle inequality that

$$d(x, y) \geq d(x, x') - d(x', y).$$

We also have

$$d(x, x') - d(x', y) > d(x, x') - (d(x, x') - \varepsilon) = \varepsilon.$$

Therefore,

$$d(x, y) > \varepsilon,$$

and thus, $B(x', \gamma) \subseteq \overline{B}(x, \varepsilon)^c$. Therefore, for all $x \in \overline{B}(x, \varepsilon)^c$, there exists an open set $x' \in B(x', \gamma) \subseteq \overline{B}(x, \varepsilon)^c$, and thus $\overline{B}(x, \varepsilon)$ is closed.

Proposition 1.3

Let $(X, \mathcal{T})$ be a topological space. Then

1. $X, \emptyset$ are closed;
2. Finite union of closed sets are closed;
3. Intersections of closed sets are closed.

Proof. Done in the homework. (1) is trivial, and (2,3) follow from De Morgan's laws. ■

Definition 1.7: Hausdorff

A topological space X is Hausdorff if for all distinct $x, y \in X$, there exist open sets U_x, U_y such that $x \in U_x$, $y \in U_y$, and $U_x \cap U_y = \emptyset$.

For example, the trivial topology on $\mathbb{R}$ is *not* Hausdorff. However, we would expect all nice spaces to be Hausdorff, such as $\mathbb{R}$ with the normal topology, so we study Hausdorff spaces in detail.

Here's another example interesting. Let $X = \mathbb{R} \cup \{0'\}$, where $0'$ is some formal point outside of $\mathbb{R}$. Every open set not containing 0 in the standard topology on $\mathbb{R}$ is open, and additionally for all open sets U containing 0, we instead include the new open set U' such that $U' = U \cup \{0'\}$. This is basically just $X = \mathbb{R} \amalg \mathbb{R}/\sim$. This space is *not Hausdorff*, since *every open set* containing 0 must also contain $0'$, so although $0 \neq 0'$, there exist no disjoint neighborhoods of these points. We usually call this a line with double 0.

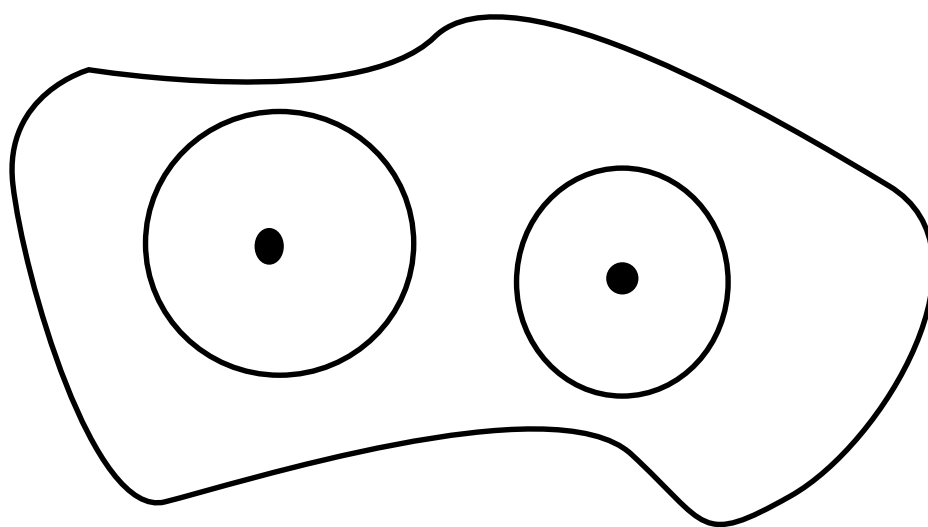


Figure 1.1: Hausdorff